



Daffodil
International
University

DEPARTMENT OF COMPUTING AND INFORMATION SYSTEM



Assignment no. 2

Course Title: Object Oriented Programming Lab

Course Code: CIS216L

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Java Basics

Problem 1: Hello World

Source Code:

```
J Hello.java > Hello
1 public class Hello {
    Run | Debug
2     public static void main(String[] args) {
3         System.out.println(x: "Hello World");
4     }
5 }
```

Output:

```
Problems Output Debug Console Terminal Ports
PS C:\Users\rizon\Desktop\java lab> & 'C:\Program Files\Java\jdk-25.0.3\bin\java.exe' '-XX:+ShowCodeDetailsInExceptionMessages' '-cp' 'C:\Users\rizon\AppData\Roaming\Code\User\workspaceStorage\6625ad99ca2b8c86295704d7183693fd\redhat.java\jdt_ws\java lab_c60b4e2e\bin' 'Hello'
Hello World
PS C:\Users\rizon\Desktop\java lab>
```

Problem 2: Comments Example

Source code:

```
J CommentsExample.java > CommentsExample
1 public class CommentsExample {
    Run | Debug
2     public static void main(String[] args) {
3         // This is first comment
4         System.out.println(x: "Welcome to Java");
5
6         /*
7         This is a
8         multi-line example
9         */
10        System.out.println(x: "This is a multi-line comment example");
11    }
12 }
```

Output:

```
Problems Output ... Filter (e.g. text, excludeText, t...) Code
[Running] cd "c:\Users\rizon\Desktop\java lab\" && javac CommentsExample.java && java
CommentsExample
Welcome to Java
This is a multi-line comment example

[Done] exited with code=0 in 2.608 seconds
```

Problem 3: Data type demo

Source code:

```
J DataTypesDemo.java > ...
1 public class DataTypesDemo {
  Run | Debug
2   public static void main(String[] args) {
3       byte age = 20;
4       short distance = 1500;    The value of the local variable distance is not used
5       int salary = 50000;
6       long population = 8000000000L;
7
8       float height = 5.7f;
9       double gpa = 3.50;
10
11      char grade = 'A';
12      boolean isPassed = true;
13
14      String studentName = "Rizon";
15      int[] testScores = {95, 85, 88};
16
17      System.out.println(x: "--- Student Details ---");
18      System.out.println("Name: " + studentName);
19      System.out.println("Age: " + age + " years");
20      System.out.println("Height: " + height + " ft");
21      System.out.println("GPA: " + gpa);
22      System.out.println("Grade: " + grade);
23      System.out.println("Passed Course: " + isPassed);
24
25      System.out.println(x: "\n--- Additional Info ---");
26      System.out.println("Monthly Salary: $" + salary);
27      System.out.println("City Population: " + population);
28
29      System.out.print(s: "Test Scores: ");
30      for (int score : testScores) {
31          System.out.print(score + " ");
32      }
33      System.out.println();
34  }
35 }
```

Output:

```
PS C:\Users\rizon\Desktop\java lab> & 'C:\ProgramData\Roaming\Code\User\workspaceStorage\6625
--- Student Details ---
Name: Rizon
Age: 20 years
Height: 5.7 ft
GPA: 3.5
Grade: A
Passed Course: true

--- Additional Info ---
Monthly Salary: $50000
City Population: 8000000000
Test Scores: 95 85 88
PS C:\Users\rizon\Desktop\java lab>
```

Problem 4: Variable demo

Source code:

```

1 public class VariablesDemo {
2     int studentAge = 20;
3     static String schoolName = "Daffodil International University";
4
5     public void displayDetails() {
6         int score = 95;
7         final double MIN_PASSING_GRADE = 3.0;
8
9         System.out.println(x: "Student Record");
10        System.out.println("Student Age (Instance): " + studentAge);
11        System.out.println("School (Static): " + schoolName);
12        System.out.println("Exam Score (Local): " + score);
13        System.out.println("Passing Grade (Final): " + MIN_PASSING_GRADE);
14    }
15
16    Run | Debug
17    public static void main(String[] args) {
18        VariablesDemo student1 = new VariablesDemo();
19        student1.displayDetails();
20
21        System.out.println(x: "\nStatic Access");
22        System.out.println("School Name via Class: " + VariablesDemo.schoolName);
23    }
24 }

```

Output:

```
Student Record
Student Age (Instance): 20
School (Static): Daffodil International University
Exam Score (Local): 95
Passing Grade (Final): 3.0

Static Access
School Name via Class: Daffodil International University

[Done] exited with code=0 in 0.601 seconds
```

Problem 5

Source code:

```
J SimpleOperatorsDemo.java > SimpleOperatorsDemo
1  public class SimpleOperatorsDemo {
    Run | Debug
2      public static void main(String[] args) {
3          int a = 10, b = 3;
4
5          System.out.println("a + b = " + (a + b));
6          System.out.println("a - b = " + (a - b));
7
8          System.out.println("a > b ? " + (a > b));
9
10         boolean x = true, y = false;
11         System.out.println("x && y = " + (x && y));
12
13         a += 5;
14         System.out.println("a after += 5: " + a);
15
16         int max = (a > b) ? a : b;
17         System.out.println("Maximum = " + max);
18     }
19 }
20
```

Output:

```
a + b = 13
a - b = 7
a > b ? true
x && y = false
a after += 5: 15
Maximum = 15
PS C:\Users\rizon\Desktop\java lab>
```

Problem 6 : Decision Making Demo

Source Code:

```
DecisionMakingDemo.java > DecisionMakingDemo
1  public class DecisionMakingDemo {
2      Run | Debug
3      public static void main(String[] args) {
4          int number = 10;
5          if (number > 0) {
6              System.out.println(x: "The number is positive.");
7          }
8
9          if (number % 2 == 0) {
10             System.out.println(x: "The number is even.");
11         } else {
12             System.out.println(x: "The number is odd.");
13         }
14
15         if (number < 0) {
16             System.out.println(x: "The number is negative.");
17         } else if (number == 0) {
18             System.out.println(x: "The number is zero.");
19         } else {
20             System.out.println(x: "The number is positive.");
21         }
22
23         int day = 3;
24         switch (day) {
25             case 1:
26                 System.out.println(x: "Monday");
27                 break;
28             case 2:
29                 System.out.println(x: "Tuesday");
30                 break;
31             case 3:
32                 System.out.println(x: "Wednesday");
33                 break;
34             default:
35                 System.out.println(x: "Other day");
36                 break;
37         }
38     }
39 }
```

Output:

```
The number is positive.
The number is even.
The number is positive.
Wednesday
PS C:\Users\rizon\Desktop\java lab>
```

Problem 7: Loops demo

Source Code:

```
J LoopsDemo.java > LoopsDemo > main(String[])
1 public class LoopsDemo {
    Run | Debug
2     public static void main(String[] args) {
3         // For Loop
4         System.out.println(x: "1. For Loop");
5         for (int i = 1; i <= 5; i++) {
6             System.out.println("Count: " + i);
7         }
8
9         // While Loop
10        System.out.println(x: "\n2. While Loop");
11        int j = 1;
12        while (j <= 5) {
13            System.out.println("Count: " + j);
14            j++;
15        }
16
17        // Do-While Loop
18        System.out.println(x: "\n3. Do-While Loop");
19        int k = 1;
20        do {
21            System.out.println("Count: " + k);
22            k++;
23        } while (k <= 5);
24
25        // Enhanced For Loop
26        System.out.println(x: "\n4. Enhanced For Loop");
27        int[] scores = {10, 20, 30, 40, 50};
28        for (int score : scores) {
29            System.out.println("Score: " + score);
30        }
31    }
32 }
```

Output:

```
Count: 1
Count: 2
Count: 3
Count: 4
Count: 5

3. Do-While Loop
Count: 1
Count: 2
Count: 3
Count: 4
Count: 5

4. Enhanced For Loop
Score: 10
Score: 20
Score: 30
Score: 40
Score: 50
PS C:\Users\rizon\Desktop\java lab>
```